



Foundations of Programming

Assignment No.: 22

PROBLEM STATEMENT: Write a menu driven program in C to perform various string operations using in-built functions.

OBJECTIVE:

- To understand the concept of **strings in the C programming language**.
- To learn how to use **built-in string functions** for string manipulation.
- To perform operations such as **length calculation, copying, concatenation, comparison, and reversal** using menu options.

THEORY:

In the C programming language, a **string** is a sequence of characters terminated by a null character (`'\0'`). Strings are stored in **character arrays** and manipulated using functions available in the **string.h header file**.

Common **built-in string functions** include `strlen()` for finding the length of a string, `strcpy()` for copying one string to another, `strcat()` for concatenating two strings, and `strcmp()` for comparing two strings.

A **menu driven program** allows the user to select different operations from a list of options. Based on the user's choice, the corresponding string function is executed. This approach makes the program interactive and easy to use.

PLATFORM: Linux

INPUT: Two strings entered by the user.

```

1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char str1[100], str2[100], result[200];
6      int choice;
7
8      printf("Enter first string: ");
9      scanf("%s", str1);
10
11     printf("Enter second string: ");
12     scanf("%s", str2);
13
14     printf("\nMENU:\n");
15     printf("1. Length of strings\n");
16     printf("2. Copy string\n");
17     printf("3. Concatenate strings\n");
18     printf("4. Compare strings\n");
19
20     printf("Choice: ");
21     scanf("%d", &choice);
22
23     switch(choice) {
24         case 1:
25             printf("Length of first string = %lu\n", strlen(str1));
26             printf("Length of second string = %lu\n", strlen(str2));
27             break;
28
29         case 2:
30             strcpy(result, str1);
31             printf("Copied string: %s\n", result);
32             break;
33
34         case 3:
35             strcpy(result, str1);
36             strcat(result, str2);
37             printf("Concatenated string: %s\n", result);
38             break;
39
40         case 4:
41             if(strcmp(str1, str2) == 0)
42                 printf("Strings are equal\n");
43             else
44                 printf("Strings are not equal\n");
45             break;
46
47         default:
48             printf("Invalid choice\n");
49     }
50
51     return 0;
52 }

```

OUTPUT: Displays results of different string operations such as **length, copy, concatenation, and comparison.**

```
Enter first string: Hello
Enter second string: World

MENU:
1. Length of strings
2. Copy string
3. Concatenate strings
4. Compare strings
Choice: 3
Concatenated string: HelloWorld
```

SAMPLE INPUT:

Enter first string: Hello
Enter second string: World
Choice: 3

SAMPLE OUTPUT:

Concatenated string: HelloWorld

CONCLUSION: Thus, the C program successfully demonstrates **different string operations using built-in functions** through a **menu driven approach**, making the program interactive and easy to use.

FAQs:

1. What are **built-in string functions** in C and why are they used?
2. What is the purpose of the **string.h header file** in C programs?
3. How does a **menu driven program improve user interaction** in programming?
- 4.